

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 3.2: Probability I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.2: Probability I
Driving Question	DRIVING QUESTION: If past outcomes of a chance event — like a coin toss or a football match — do not change future probabilities, how can we use mathematics to predict what is likely to happen, and why does this understanding protect us from the dangers of gambling?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Equally Likely Events: Is Gor Mahia "Due" for a Loss?	Teacher shows learners a sequence of 10 coin-toss results (e.g., H H H H H H H H H) projected on the board and asks: 'Gor Mahia has won their last 8 home matches — are they now *due* for a loss?' Learners record their gut-feeling predictions and share with a shoulder partner. Teacher cold-calls 4–5 learners (wait time: 10 seconds) to surface the common misconception that a long winning streak makes a loss more likely. Teacher then physically tosses a coin 10 times in front of the class, recording each result; learners tally and note that each toss is unaffected by the previous one.	Each trial of a chance event — a coin toss, a dice roll, a football match outcome — is an **independent event**: the result of one trial has absolutely no effect on the next. The coin has no memory; neither does a football pitch. This directly dismantles the 'gambler's fallacy' — the dangerous belief that past outcomes shift future probabilities. Teachers should emphasise this concept repeatedly throughout the unit, as it is the conceptual anchor for all eight lessons and the primary anti-gambling safeguard embedded in the curriculum.	Equally likely outcomes are outcomes that have exactly the same chance of occurring. For a fair coin: {Head, Tail} — both outcomes are equally likely, so each has probability 1/2. For a fair six-sided die: {1, 2, 3, 4, 5, 6} — all six outcomes are equally likely, each with probability 1/6. The sample space S is the complete set of all possible outcomes of an experiment. Teachers should write $S = \{H, T\}$ and $S = \{1, 2, 3, 4, 5, 6\}$ explicitly on the board and have learners copy these as reference examples. Pedagogical note: resist moving to the formula in this lesson — the goal is conceptual grounding in equally likely outcomes and independence before formalising $P(\text{event})$.
Lesson 2 Computing Probability of Equally Likely Outcomes	Teacher distributes coloured counters (or draws a bag on the board containing 3 red, 2 blue, and 1 yellow counter — representing the colours of a Kenyan flag-themed lucky-draw bag at a school fundraiser in Nakuru). Learners predict which colour is most likely to be drawn. Teacher conducts 12 draws (with	The theoretical probability formula is: **$P(\text{event}) = \text{number of favourable outcomes} \div \text{total number of equally likely outcomes}$**. This formula gives the *expected* long-run proportion of times an event will occur, assuming all outcomes are equally likely. Key numerical facts: $P(\text{red counter}) = 3/6 = 1/2$; $P(\text{blue}) = 2/6 = 1/3$;	To apply the formula correctly: (1) List the full sample space S; (2) Count total outcomes $n(S)$; (3) Identify and count favourable outcomes $n(E)$; (4) Compute $P(E) = n(E) \div n(S)$. Example: A standard deck of 52 playing cards — $P(\text{drawing a King}) = 4/52 = 1/13$. Kenya-context example: A matatu route number is chosen

	replacement, shaking the bag each time) and learners tally results on a frequency table. Teacher cold-calls 3 learners (wait time: 8 seconds) to compare their tally-based (experimental) fractions with their initial predictions before introducing the theoretical formula.	$P(\text{yellow}) = 1/6$. Teachers should stress that the formula requires equally likely outcomes — it cannot be applied blindly to situations where outcomes are not equally likely (e.g., assuming a football match has equally likely win/draw/loss outcomes is a common misapplication). Probabilities are always expressed as fractions, decimals, or percentages.	at random from Nairobi CBD routes {23, 33, 34, 46, 58, 111, 125} — $P(\text{route number} > 50) = 3/7$. Teachers should work through at least two full examples on the board using the four-step method before setting learners to practise in pairs. Warn learners: if they cannot list the sample space, they cannot use this formula.
Lesson 3 Range of Probability: From Impossible to Certain	Teacher writes five events on the board and asks learners to silently rank them from 'absolutely cannot happen' to 'must happen': (a) It snows in Kisumu in July; (b) The sun rises tomorrow over Mount Kenya; (c) A fair coin lands on Heads; (d) A student chosen at random from a Grade 10 class in Eldoret is under 30 years old; (e) Rolling a 7 on a standard six-sided die. Learners write their rankings individually (2 minutes), then compare with a partner. Teacher cold-calls 4 learners (wait time: 10 seconds) and records disagreements on the board to motivate the need for a precise numerical scale.	The probability of any event E always satisfies $0 \leq P(E) \leq 1$. $P(E) = 0$ means the event is 'impossible' — it can never occur (e.g., rolling a 7 on a six-sided die; snow in Kisumu in July). $P(E) = 1$ means the event is 'certain' — it must always occur (e.g., the sun rising tomorrow; a learner in Grade 10 being younger than 100 years old). Values between 0 and 1 express degrees of likelihood. Teachers should draw the probability scale (number line from 0 to 1) on the board and have learners physically place each event from the opening activity on the scale. This visual is a powerful tool for correcting intuitive errors.	Key probability benchmarks on the scale: 0 (impossible) → 0.25 (unlikely) → 0.5 (equally likely as unlikely) → 0.75 (likely) → 1 (certain). The complement rule follows directly: $P(E \text{ does NOT occur}) = 1 - P(E)$. Example: If $P(\text{Gor Mahia wins a match}) = 0.6$, then $P(\text{Gor Mahia does NOT win}) = 1 - 0.6 = 0.4$. Kenya-context example: Weather records show it rains on about 120 days per year in Kericho (the tea capital). $P(\text{rain on a randomly chosen day}) \approx 120/365 \approx 0.33$. Pedagogical note: the complement rule is foundational for Lessons 4–7; ensure all learners can state and apply it before proceeding.
Lesson 4 Mutually Exclusive Events — Identification & The Addition Rule	Teacher presents the scenario: 'A matatu leaves Nairobi Bus Station. It can go to Mombasa, Kisumu, or Nakuru — but it can only take one route at a time.' Teacher asks: 'Can the matatu arrive in Mombasa AND Kisumu on the same trip?' Learners discuss in groups of three for 3 minutes. Teacher then shows a Venn diagram with two non-overlapping circles (labelled 'Wins' and 'Draws' for a single football match) and one with overlapping circles (labelled 'Learner studies Maths' and 'Learner studies Biology') and asks learners which diagram represents mutually exclusive events. Cold-call 4 learners (wait time: 8 seconds).	Two events A and B are 'mutually exclusive' if they cannot both occur on the same trial — they share no outcomes, so $P(A \text{ and } B) = 0$. The Venn diagram for mutually exclusive events shows two non-overlapping circles. The 'Addition Rule for mutually exclusive events' is: $P(A \text{ or } B) = P(A) + P(B)$. Kenya-context example: On a single fair die roll, $A = \{\text{rolling a 2}\}$ and $B = \{\text{rolling a 5}\}$ are mutually exclusive. $P(A \text{ or } B) = 1/6 + 1/6 = 2/6 = 1/3$. Non-mutually-exclusive example (for contrast): 'Learner passes Maths' and 'Learner passes Kiswahili' can both occur simultaneously — these are NOT mutually exclusive and the addition rule above does NOT apply. Teachers must spend adequate time on identification before the rule.	Step-by-step procedure for the Addition Rule: (1) Check whether A and B are mutually exclusive — confirm they cannot happen together; (2) If yes, apply $P(A \text{ or } B) = P(A) + P(B)$; (3) Verify the answer is ≤ 1 . Extended Kenya example: A spinner has 8 equal sectors: 3 green (for Gor Mahia), 2 red (for AFC Leopards), 2 yellow, 1 blue. $P(\text{green or red}) = 3/8 + 2/8 = 5/8$. The Addition Rule extends to three or more mutually exclusive events: $P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C)$. Pedagogical note: a frequent error is learners adding probabilities for events that are NOT mutually exclusive — address this explicitly using a counterexample before learners practise independently.
Lesson 5 Experimental Probability & The Law of Large	Teacher organises learners into groups of four. Each group tosses a coin 20 times and records results in a tally table (H or T).	'Experimental (relative frequency) probability' is calculated as: $P(\text{event}) \approx \text{number of times event occurred} \div \text{total}$	Experimental probability is essential when theoretical equally-likely outcomes cannot be assumed — e.g., estimating the

Numbers	Groups report their experimental proportion of Heads to the class; teacher compiles all groups' results on the board, progressively computing the cumulative proportion as more tosses are added (20, 40, 60, 80... up to the full class total). Learners observe on a projected graph (or hand-drawn axes) how the cumulative proportion of Heads fluctuates widely at first then gradually approaches 0.5 as the number of trials increases. Cold-call 3 learners (wait time: 10 seconds): 'What do you notice about the proportion as the number of tosses grows?'	number of trials**. This is an *estimate* of probability based on observed data, not a calculated certainty. The **Law of Large Numbers** states that as the number of trials increases, the experimental probability gets closer and closer to the theoretical probability. With only 10 tosses, results can vary wildly (e.g., 7 Heads out of 10 = 0.7, far from 0.5); with 1 000 tosses, the proportion will be very close to 0.5. Teachers should connect this explicitly to gambling: a casino runs millions of games — the Law of Large Numbers guarantees the house always collects its mathematical edge in the long run, even if individual gamblers occasionally win.	probability that a maize crop in Kitale will be affected by drought in a given year, or estimating the probability that a matatu arrives on time. Worked example: In a survey of 200 customers at a githeri stall in Gikomba Market, 80 ordered extra soup. Experimental $P(\text{customer orders extra soup}) = 80/200 = 0.4$. This does NOT mean exactly 2 in every 5 customers will order soup — it is an estimate that improves with more data. Pedagogical note: explicitly contrast experimental and theoretical probability using the coin-toss data just collected; ensure learners understand that neither is 'wrong' — they serve different purposes.
Lesson 6 Combined Events & Sample Spaces Using Tables and Tree Diagrams	Teacher poses the scenario: 'Mama Wanjiku is making a packed lunch. She randomly picks one starch (ugali or rice) and one protein (beef, chicken, or beans — nyama ya ng'ombe, kuku, au maharagwe). How many different lunch combinations are possible, and what is the probability she picks ugali with beans?' Learners sketch their own diagrams individually for 3 minutes before teacher reveals a systematic approach. Cold-call 4 learners to share their methods (wait time: 8 seconds); expect a mix of guesses, lists, and diagrams to surface the need for systematic methods.	When two or more choices are made independently, the **sample space of combined events** can be mapped systematically using: (1) A **two-way table (grid)** — columns represent one variable, rows the other, and cells represent combined outcomes; (2) A **tree diagram** — branches represent sequential choices; multiply along branches for combined probability. For Mama Wanjiku's lunch: total combinations = $2 \times 3 = 6$. The **Fundamental Counting Principle** states: if event 1 has m outcomes and event 2 has n outcomes, the combined experiment has $m \times n$ outcomes. $P(\text{ugali AND beans}) = 1/6$. Teachers should construct both the table and tree diagram on the board for the same example so learners see the equivalence.	Tree diagram procedure: (1) Draw a branch for each outcome of the first event; (2) From each branch, draw sub-branches for each outcome of the second event; (3) List all end-points — these are the sample space; (4) Each complete path has probability = product of individual branch probabilities (for independent events). Kenya exam-style example: A school in Thika randomly selects a learner's house (Lions, Eagles, or Cheetahs) and a uniform colour (white or blue). $P(\text{Lions house AND blue uniform}) = 1/3 \times 1/2 = 1/6$. Pedagogical note: tree diagrams are also the ideal visual scaffold for Lesson 7's Multiplication Rule — ensure learners are comfortable constructing them before that lesson.
Lesson 7 Probability of Independent Events in Context: The Multiplication Rule	Teacher returns to the driving question anchoring phenomenon: 'Gor Mahia plays two separate football matches — one in Nairobi and one in Kisumu — on different weekends. If $P(\text{win each match}) = 0.6$, what is the probability they win BOTH matches?' Learners first attempt the problem individually using a tree diagram from Lesson 6 (3 minutes), then share answers with a partner. Teacher cold-calls 3 learners (wait time: 10 seconds) and records different answers on the board (expect	Two events A and B are **independent** if the occurrence of A has no effect on the probability of B — and vice versa. This is the mathematical formalisation of the independence concept introduced in Lesson 1. The **Multiplication Rule for independent events** is: $P(A \text{ and } B) = P(A) \times P(B)$. Gor Mahia example: $P(\text{win match 1 AND win match 2}) = 0.6 \times 0.6 = 0.36$. Important: the multiplication rule only applies when events are truly independent — teachers must always prompt learners to	The Multiplication Rule extends to three or more independent events: $P(A \text{ and } B \text{ and } C) = P(A) \times P(B) \times P(C)$. Kenya-context worked example: A farmer in Meru plants three separate plots. The probability that each plot produces a good harvest is 0.7 (independent due to separate soils and irrigation). $P(\text{all three plots produce good harvests}) = 0.7 \times 0.7 \times 0.7 = 0.343$. Contrast with the Addition Rule: Addition Rule → 'OR' → mutually exclusive events → ADD probabilities. Multiplication Rule →

	some to add, some to multiply, some to average). Teacher uses the tree diagram to confirm the correct approach, connecting branch-multiplication to the formal rule.	justify independence before applying the formula. Gambling connection: a gambler who has won three bets in a row is NOT more likely to win the fourth — each bet is independent, and $P(\text{win bet 4})$ is unchanged.	'AND' → independent events → MULTIPLY probabilities. Pedagogical note: the OR/AND → ADD/MULTIPLY mnemonic is a powerful memory aid; write it prominently on the board and revisit it in Lesson 8.
Lesson 8 Consolidation, Final Explanation & DQB Closure: Why Mathematics — Not Luck — Governs the Odds	Teacher projects the original Driving Question on the board and returns to the Lesson 1 anchoring phenomenon: 'Gor Mahia has now won 12 matches in a row. Are they due for a loss?' Learners revisit their Lesson 1 prediction cards and write a revised answer using mathematical language and probability values (5 minutes individual writing). Teacher facilitates a structured DQB closure: each learner's sticky note question from the DQB is read aloud; the class collectively identifies which lesson answered it and how. Teacher cold-calls 5 learners (wait time: 10 seconds each) to present one probability concept and its real-world anti-gambling implication. Class constructs a final collective summary model.	Every chance event studied in this unit — coin tosses, dice rolls, counter draws, football matches, farm harvests — is governed by fixed mathematical probabilities that do not change based on past outcomes. The **gambler's fallacy** (the belief that a run of outcomes makes the opposite outcome 'due') is mathematically false and financially dangerous. The Law of Large Numbers guarantees that over many trials, the house's mathematical edge accumulates — meaning gambling is a losing strategy in the long run, regardless of short-term wins. The full conceptual map of Probability I: Sample Space → Theoretical Probability ($P = n(E)/n(S)$) → Probability Scale (0 to 1) → Complement Rule → Mutually Exclusive Events (Addition Rule) → Experimental Probability → Combined Events (Tables & Trees) → Independent Events (Multiplication Rule).	Final synthesis of all eight lessons' rules in one consolidated reference: (1) $P(E) = n(E)/n(S)$ for equally likely outcomes; (2) $0 \leq P(E) \leq 1$ always; (3) $P(E') = 1 - P(E)$ [Complement Rule]; (4) If A and B are mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$ [Addition Rule]; (5) Experimental $P(E) \approx \text{frequency/total trials}$ → approaches theoretical $P(E)$ as trials increase [Law of Large Numbers]; (6) Combined sample spaces: use tables or tree diagrams; total outcomes = $m \times n$ [Counting Principle]; (7) If A and B are independent: $P(A \text{ and } B) = P(A) \times P(B)$ [Multiplication Rule]. Pedagogical note: distribute or display this consolidated rule sheet as a revision reference. Emphasise the life-skill message: understanding these rules mathematically empowers learners to recognise and reject the false logic that underlies gambling addiction.

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